Causal Inference with LLMs

Day 1: Foundations & Measurement

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instats Seminar

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Day 1: Foundations & Measurement

- Foundations of Causal Inference
- Graphical Models and Identification
- Measurement and LLMs

Interactive App

- **URL:** Click here to open the Interactive Streamlit App
- **No Installation:** Runs entirely in the browser via Streamlit.
- **Navigation:** Use the sidebar to switch between workshop days.
- **Step-by-Step:** Each page walks through core computations.
- **Consistency:** Uses the U.S. Congressional speeches dataset (Hein Bound 111); results match local scripts.

The Causal Framework

Central question: What is the effect of an intervention?

Basic terms:

- **Unit:** an individual entity of analysis (e.g., person, document, organization)
- **Treatment:** an action, manipulation, exposure, or condition applied to a unit
- **Intervention:** the act of assigning a unit to a treatment
- $D \in \{0, 1\}$: treatment indicator ($1 =$ unit receives treatment, $0 =$ does not)
- Y: outcome variable (measured response of the unit)
- X: vector of observed covariates (pre-treatment characteristics of the unit)

Observed comparison:

$$\mathbb{E}[Y|D = 1] - \mathbb{E}[Y|D = 0]$$

Easy to compute, but not necessarily causal

Foundations of Causal Inference (cont.)

Expectation:

$$\mathbb{E}[Y]$$

- Average (mean) value of Y across all units

Conditional expectation:

$$\mathbb{E}[Y|D = d]$$

- Average value of Y among units with treatment status $D = d$

Issue:

- Observational differences combine causal effects and selection bias
- Units receiving treatment differ systematically in their attributes (covariates X)

Goal: isolate the causal component

The Fundamental Problem of Causal Inference

For each unit i, define:

$$Y_i(1), \quad Y_i(0)$$

Interpretation:

- $Y_i(1)$: outcome for unit i if it receives the treatment
- $Y_i(0)$: outcome for unit i if it does not receive the treatment
- The values 1 and 0 index the two possible treatment states

Observed outcome:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$$

Counterfactuals

Counterfactual:

- The outcome that would have occurred had the unit received the other treatment state

Example:

- If $D_i = 1$, we observe $Y_i(1)$ and the counterfactual is $Y_i(0)$
- If $D_i = 0$, we observe $Y_i(0)$ and the counterfactual is $Y_i(1)$

Key limitation:

- Only one potential outcome is observed for each unit
- Equivalently, the counterfactual outcome is fundamentally unobservable

Average Treatment Effects

Definition (ATE):

- Average effect of the treatment across all units in the population

$$\tau = \mathbb{E}[Y(1) - Y(0)]$$

Definition (ATT):

- Average causal effect among units that actually received the treatment

$$\tau_{\text{ATT}} = \mathbb{E}[Y(1) - Y(0) \mid D = 1]$$

Challenge:

- Neither is directly observable, since $Y(1)$ and $Y(0)$ are never simultaneously observed for the same unit — identification assumptions make them estimable from data

Naive Estimator and Bias

Naive estimator:

- Difference in average observed outcomes between treated and non-treated units, called 'control units'

$$\mathbb{E}[Y|D=1] - \mathbb{E}[Y|D=0]$$

Decomposition:

$$\underbrace{\mathbb{E}[Y|D=1] - \mathbb{E}[Y|D=0]}_{\text{naive estimator}} =$$

$$\mathbb{E}[Y(1)|D=1] - \mathbb{E}[Y(0)|D=0] + \mathbb{E}[Y(0)|D=1] - \mathbb{E}[Y(0)|D=1] =$$

$$\underbrace{\mathbb{E}[Y(1)|D=1] - \mathbb{E}[Y(0)|D=1]}_{\tau\ (\text{ATT})} + \underbrace{\mathbb{E}[Y(0)|D=1] - \mathbb{E}[Y(0)|D=0]}_{\text{selection bias}}$$

Source of bias:

- The bias is zero if and only if $\mathbb{E}[Y(0)|D=1] = \mathbb{E}[Y(0)|D=0]$, that is, if treated and control units would have had the same average outcome in the absence of treatment

Conditioning on Covariates

$$\mathbb{E}[Y \mid X = x]$$

Interpretation:

- Average outcome among units sharing the same covariate value $X = x$
- Restricts the comparison to a homogeneous subgroup

Key idea:

- Controlling for X means comparing treated and control units *within* the same value of X
- This removes differences in outcomes that are due to differences in X, not in the treatment assignment D

Conditional Independence

Notation:

$$A \perp B \mid C$$

- A is independent of B given C
- Knowing B provides no additional information about A once C is known

In causal inference: ignorability (unconfoundedness)

$$(Y(1), Y(0)) \perp D \mid X$$

Interpretation: If, given X, the potential outcomes are independent of treatment assignment, then:

- Within each value of X, treatment assignment is as good as random

SUTVA

Stable Unit Treatment Value Assumption (SUTVA)

Components:

- **No interference:**
  - The outcome of unit i depends only on its own treatment D_i
  - Formally: $Y_i(D) = Y_i(D_i)$, where D is the full vector of treatment assignments across all units
- **No hidden versions of treatment:**
  - The treatment $D = 1$ represents a single, well-defined intervention
  - Different implementations of the treatment do not lead to different outcomes

Implication:

- Potential outcomes $Y_i(1)$ and $Y_i(0)$ are well-defined for each unit

Role in causal inference:

- Ensures the validity of the potential outcomes notation and the derivation of the identification results that follow

Ignorability

Ignorability (Unconfoundedness):

$$(Y(1), Y(0)) \perp D \mid X$$

Confounder:

- A variable that causally affects both D and Y, creating a spurious association between them
- Ignorability requires that all confounders are observed and measured as elements of X

Implication:

- The selection bias term vanishes ($\mathbb{E}[Y(0) \mid D = 1, X] = \mathbb{E}[Y(0) \mid D = 0, X]$)
- Within levels of X, differences in outcomes between $D = 1$ and $D = 0$ can be attributed to the treatment

Limitation:

- Ignorability–that all confounders are observed and measured as elements of X–cannot be verified from observed data alone

Overlap (Positivity)

$$0 < P(D = 1|X = x) < 1$$

Notation:

- $P(D = 1|X = x)$: probability that a unit with covariates $X = x$ receives the treatment

Meaning:

- For every value x of X, both treatment states are possible
- No subgroup is deterministically treated or untreated

Role:

- Ensures comparability between treated and control units within levels of X
- If positivity fails for some x, the causal effect cannot be identified for units with $X = x$

Identification of the ATE

The adjustment formula:

- Under SUTVA, ignorability, and overlap, τ is identified and equals:

$$\tau = \mathbb{E}_X \left[\mathbb{E}[Y|D=1,X] - \mathbb{E}[Y|D=0,X] \right]$$

where $\mathbb{E}_X[\cdot]$: expectation taken over the distribution of covariates X

Interpretation:

- First compare treated and control units with the same X
- Then average these differences over all X

Key idea:

- Conditioning on X removes confounding bias under ignorability and overlap

Structural Causal Models and DAGs

Structural Causal Model (SCM):

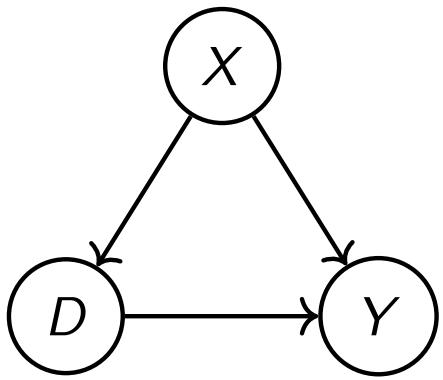
- A system of variables linked by structural (causal) relationships

Directed Acyclic Graph (DAG):

- **Nodes:** variables (e.g., D, Y, X)
- **Directed edges:** causal effects (arrows indicate direction of influence)
- **Acyclic:** no directed cycles (no variable causes itself)

Purpose:

- Encode assumptions about causal structure
- Identify which variables to condition on



X: confounder D: treatment
Y: outcome

Confounding in DAGs

Backdoor path:

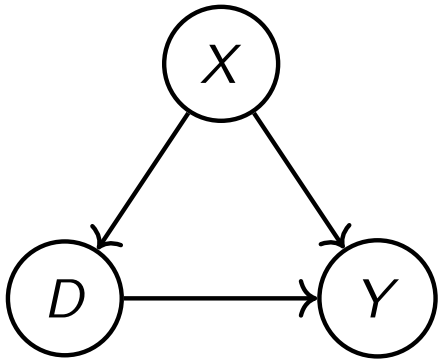
- A **path** is a sequence of variables connected by edges, ignoring arrow direction
- A path between D and Y is a **backdoor path** if its first edge has an arrow pointing into D
- Example: $D \leftarrow X \rightarrow Y$
- More generally: $D \leftarrow X_1 \rightarrow X_2 \rightarrow \cdots \rightarrow Y$

Effect:

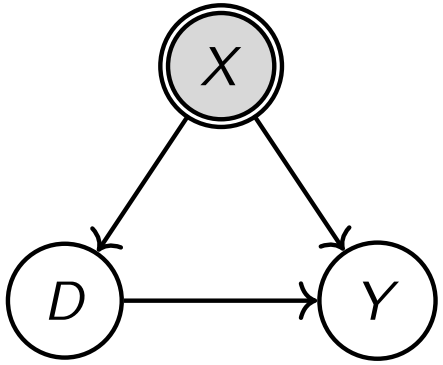
- Creates spurious association between D and Y

Solution: condition on X to strip away the spurious association it creates between D and Y

Confounded



Adjusted (condition on X)



Backdoor Criterion

Definition: A set of variables X satisfies the backdoor criterion if:

- It blocks all backdoor paths between D and Y
- It does not include descendants of D — conditioning on a descendant would block part of the causal path from D to Y, removing some of the effect we seek to measure

Terminology:

- **Blocked path:** conditioning removes association flowing along the path
- **Descendant:** a variable causally affected by D

Implication:

- If X satisfies the backdoor criterion, the adjustment formula holds and the causal effect of D on Y is identified from observed data

Why Not Condition on Descendants of D?

Setup:

- M is a **mediator**: it lies on the causal path from D to Y
- M is a descendant of D: $D \to M \to Y$

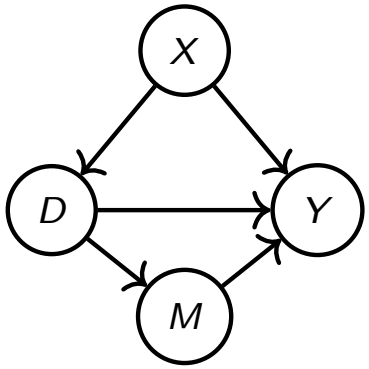
Problem:

- Conditioning on M holds fixed part of the mechanism through which D affects Y
- This blocks the indirect causal path $D \to M \to Y$
- The estimated effect of D on Y is then only the direct path $D \to Y$, not the total effect

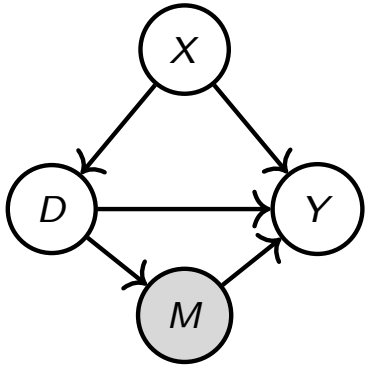
Conclusion:

- Including a descendant of D in the adjustment set leads to underestimation of the total causal effect

Correct: do not condition on M



Wrong: condition on M



M shaded = conditioned on; indirect path blocked.

Adjustment Formula

- **The identification formula that holds under the backdoor criterion**

$$P(Y|\,\mathrm{do}(D=d)) = \int P(Y|D=d,X)\,dP(X)$$

Meaning:

- Left-hand side: causal effect of intervening and setting $D = d$
- Right-hand side: estimable from observed data under the backdoor criterion

Interpretation:

- Compare treated and control units with the same X
- Average these comparisons over the distribution of X

Role:

- Translates a causal quantity into an estimable statistical expression
- Connects the DAG-based identification strategy to the adjustment formula derived earlier under ignorability and overlap

Prediction vs Causation

$$P(Y \mid D) \quad \text{vs} \quad P(Y \mid \text{do}(D))$$

Prediction:

- $P(Y \mid D)$ captures the association between D and Y in observed data
- Reflects both causal effects and confounding

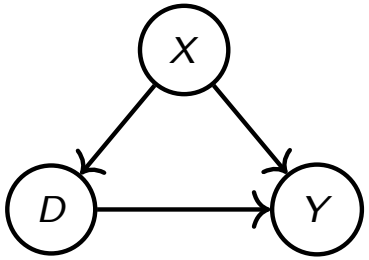
Causation:

- $P(Y \mid \text{do}(D))$ describes the effect of externally intervening to set D
- Severs the influence of any variable causing D, isolating its causal effect on Y

Why correlation does not imply causation:

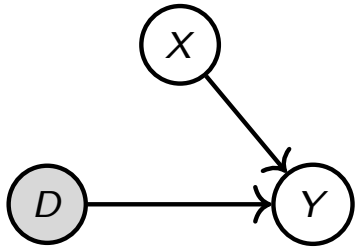
- A confounder X induces a spurious correlation between D and Y
- $P(Y \mid D = 1) \neq P(Y \mid D = 0)$ even if D has no causal effect on Y
- Yet $P(Y \mid \text{do}(D = 1)) = P(Y \mid \text{do}(D = 0))$

Observational



$D \leftarrow X \rightarrow Y$: confounded

Interventional



do(D) cuts the arrow $X \rightarrow D$

Why Causality Matters

Prediction is not enough:

- High predictive accuracy does not guarantee valid causal conclusions
- A model can exploit confounding to predict well while estimating the wrong effect

Causal reasoning is needed for:

- **Policy evaluation:** what is the effect of an intervention on Y?
- **Scientific explanation:** which variables causally drive Y, and through which mechanisms?
- **Counterfactual reasoning:** what would Y have been under a different treatment assignment?

The framework we have built:

- Potential outcomes, identification assumptions, and DAGs provide the tools to answer these questions from data

From Structured Data to Text

Classical setting:

- Variables (D, Y, X) are observed directly and measured without ambiguity

Text-based setting:

- Many variables of interest are embedded in unstructured text (documents, narratives, speech)
- They must be extracted and operationalised before causal analysis can proceed

The new challenge:

- Large Language Models (LLMs) are increasingly used to extract variables from text
- LLM-derived measurements are noisy proxies, not direct observations
- This introduces measurement error into the causal framework — the subject of what follows

Text as Data

$$W_i \in \mathscr{V}^*$$

Notation:

- W_i: text associated with unit i (a sequence of tokens)
- $\mathscr{V}$: vocabulary (set of all possible tokens)
- $\mathscr{V}^*$: all finite sequences of tokens (the asterisk denotes the Kleene star: all finite sequences, including the empty one)

Goal:

- Map W_i to structured variables (D_i, Y_i, X_i) suitable for causal analysis

LLMs as Measurement Operators

$$f_{\text{LLM}} : W_i \longrightarrow (\tilde{D}_i, \tilde{Y}_i, \tilde{X}_i)$$

Notation:

- W_i: text associated with unit i
- f_{LLM}: mapping implemented by the LLM
- $(\tilde{D}_i, \tilde{Y}_i, \tilde{X}_i)$: LLM-derived proxies of the true variables (D_i, Y_i, X_i)

Key point:

- The LLM does not observe the true variables directly — it produces noisy, potentially biased measurements
- Using $(\tilde{D}_i, \tilde{Y}_i, \tilde{X}_i)$ in place of (D_i, Y_i, X_i) introduces measurement error into the causal analysis

Examples of LLM-Constructed Variables

From text W_i, the LLM may construct:

- $\tilde{D}_i$ (treatment): presence of a policy frame, intervention, or exposure
- $\tilde{Y}_i$ (outcome): sentiment, stance, or evaluation extracted from text
- $\tilde{X}_i$ (covariates): topic, ideology, tone, or contextual features

Key point:

- The same text may contain information about treatment, outcome, and covariates simultaneously
- This creates a risk of cross-contamination: the LLM's measurement of one variable may be influenced by its reading of the others

Implication:

- The quality of causal inference depends on the quality of this measurement step
- What matters is not only random measurement error but systematic bias, which can invalidate causal estimates even at large scale

Non-Classical Measurement Error

Classical measurement error:

- Independent of true variables
- Typically assumed mean-zero and random

LLM-based measurement:

- Errors may depend on context, language, or latent features of the text
- May correlate with the true treatment D or covariates X

Implication:

- Conditioning on $\tilde{X}_i$ instead of X_i does not fully remove confounding
- The adjustment formula and ignorability may fail even when the backdoor criterion is satisfied in terms of the true variables
- Causal estimates are biased, and the bias does not vanish with larger samples

Implications and Outlook

The problem:

- Naive use of LLM-derived variables $(\tilde{D}_i, \tilde{Y}_i, \tilde{X}_i)$ in place of true variables leads to biased causal estimates
- The bias is systematic and does not vanish with larger samples

Correcting measurement error:

- Combine a small set of ground-truth labels with large-scale LLM outputs
- Use statistical correction methods to adjust for bias
- Recover valid causal estimates under appropriate conditions

Next session:

- Design-Based Supervised Learning (DSL): a principled framework for doing this

References

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Questions and Discussion

Thank you!

Questions?

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